

Geoid of Nepal from land and airborne data

Rene Forsberg, Arne V. Olesen

Oct 24, 2011

This note describes the GRAVSOFTE steps needed to do a geoid for Nepal, including both airborne and available land data from both Nepal Department of Survey and BGI (Bureau Gravimetrique, Toulouse).

The method used is remove-restore, using EGM08 augmented with GOCE to degree 360 as reference, and terrain effects from SRTM (15"). The GRAVSOFTE steps are set up as *jobs*, to run under windows. Compared to the earlier computation an enhanced theory has been used, with formal Molodensky-style continuation ("G₁-term"), computation of the quasigeoid, and subsequent transformation to the classical geoid, consistent with the use of Helmert orthometric heights.

The main program steps in the geoid computations include:

1. Computation of EGM-GOCE fields in grids (*directory EGM*, "*geocol17.job*")
2. Reduction of data for EGM and terrain (*directory TC-RD*, "*rd_air.job*" and "*rd_suface.job*")
3. Downward continuation by least-squares collocation (*directory DOWNCONT*, "*gpcol1a.job*")
4. Geoid computation by spherical FFT ("*geoid3a.job*" for quasigeoid, "*n-zeta.job*" for geoid)

The "best" geoid has been named "*nepal_geoid_oct2011.gri*" and is accessible with user-friendly interpolation software "*grid_int*".

An operational copy of the GRAVSOFTE software and associated data has been provided to the Nepal Survey Division, along with supporting software. Note that it is essential that the *path* has been set to SOFTWARE executables, and that the *job*-program, simulating unix jobs, is operational, see the GRAVSOFTE manual for details. This work has been done in connection with the visit of Kalayan Shrestha and Niraj Manandhar to Denmark for computations and training.

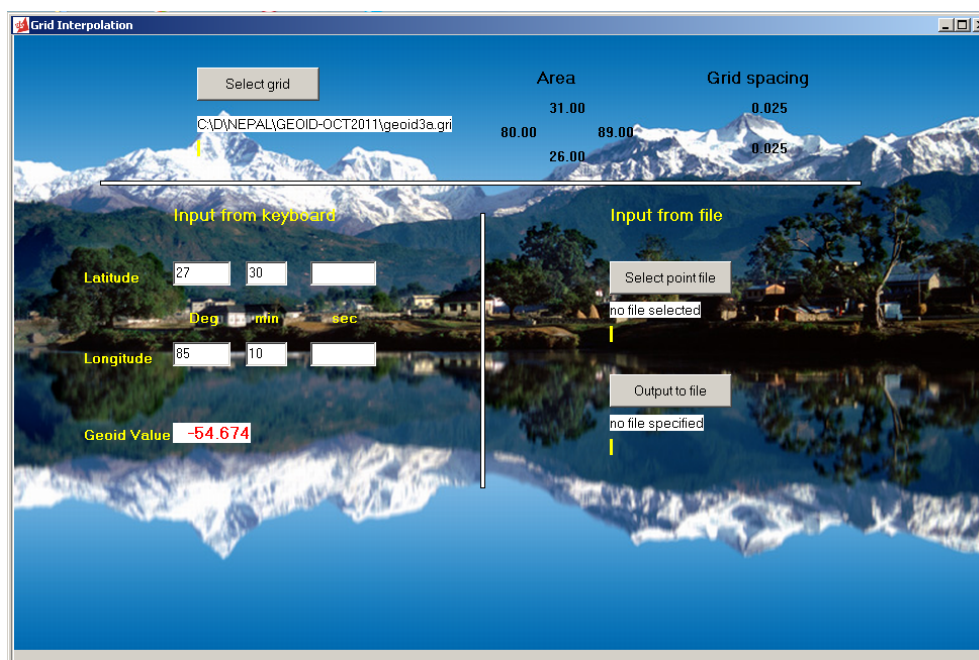


Fig. 1. "*grid_int*" software interface

Airborne data

An airborne gravity data set “*airborne4.faa*” has been used for the processing. The format of this file is *pointno*, *lat*, *lon*, *height*, *gravity (g)*, *free-air anomaly (Δg)* and *time (JD)*. It should be pointed out that this file is still *preliminary*: Chekan data have not yet been processed, and the scale-factor of the used S-38 LCR gravimeter need to be checked (risk of geoid slope if scale factor is not correct).

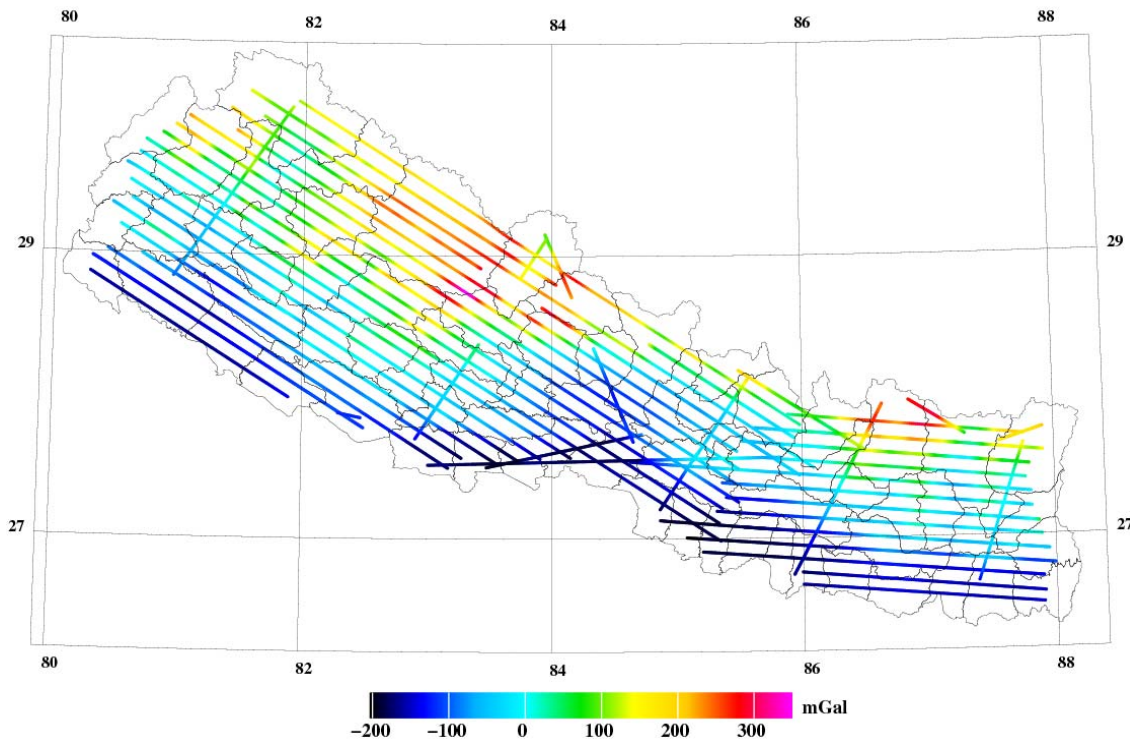


Fig. 2. Free-air anomalies at altitude(10 sec spacing data)

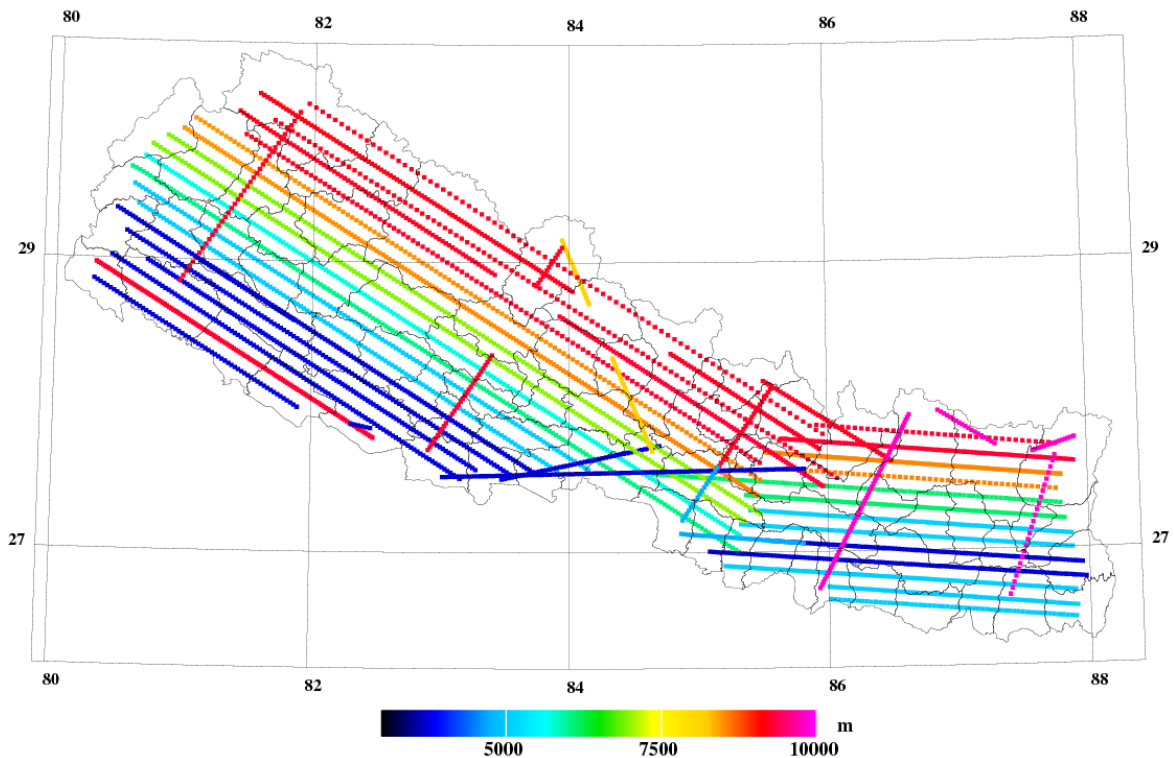


Fig.3. Height of flight tracks (30 sec spacing data)

EGM08/GOCE – computation

EGM08 supplemented with the GOCE “timewise” solution to produce an updated reference field. EGM08 coefficients in a band 90-150 is overwritten 100% by GOCE, with a linear transition in bands 80-90 and 150-160. In this way the improved medium-wavelength performance of GOCE is utilized (with program setup in “merge_goce.job”).

EGM-GOCE gravity anomalies and height anomalies are computed in “sandwich grids” at 0 and 9 km elevation. We use here remove-restore with a maximal degree of 720; this provide a maximum utilization of possible valid EGM08 data in the China and India. An constant geoid bias of 53 cm has been added to offset the output from *geocol17* and the official WGS84 EGM08 site results.

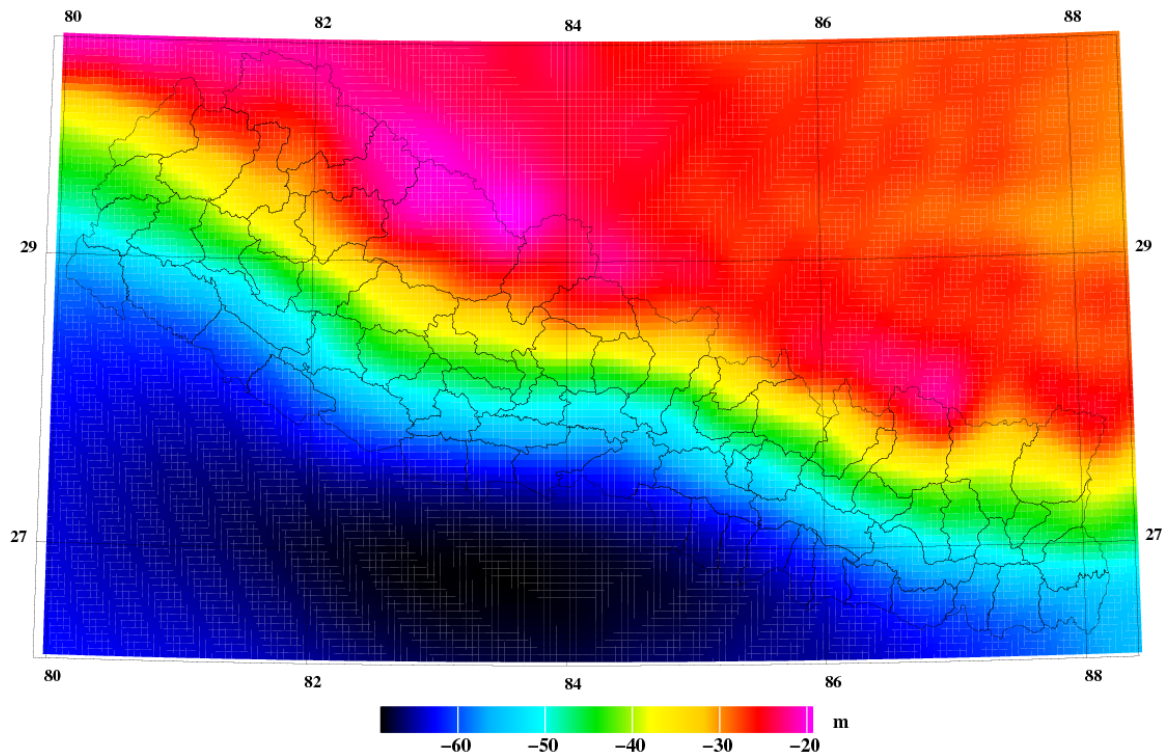


Fig. 4. EGM08-GOCE reference quasigeoid (degree 720)

Land gravity data

Land gravity data was provided by Survey Department covering both the lowlands and selected data in the high lands. The data were provided in two excel-files:

1st order gravity data.xls – 1st order control points with g-values

coordinatedata_corrected.xls – points in both lowlands and mountains with a mix of GPS and levelled heights.

The surface data with only ellipsoidal heights were converted to orthometric heights using the July 2 preliminary geoid, constrained at points with both GPS and levelling heights (GPS heights were given in local systems with 20-50 m offsets), and free-air and Bouguer anomalies in GRS80 / density 2.67 subsequently computed. These data were additionally supplemented with BGI data, mainly covering a profile from the Indian border through Kathmandu to Mt Everest (Kano, 1974 data); file name of the combined data set is “*surface.ano*”, and contain 1114 points.

All surface data were subsequently quality controlled graphically through comparison of terrain-corrected Bouguer anomalies to a terrain-corrected Bouguer anomaly grid obtained from the downward continued airborne gravity data. This QC indicated several outliers of more than 50 mgal, which were removed. A number of outliers are still present in the data set, with several more outliers and bias still likely remaining in the surface data. The corrected file “*surface_edited.faa*” contains the final surface data, including the atmosphere correction. This data set was averaged to one point pr. 3’ resolution in the final collocation downward continuation, to avoid excess weight on the surface data (file: “*surface_sel.faa*”).

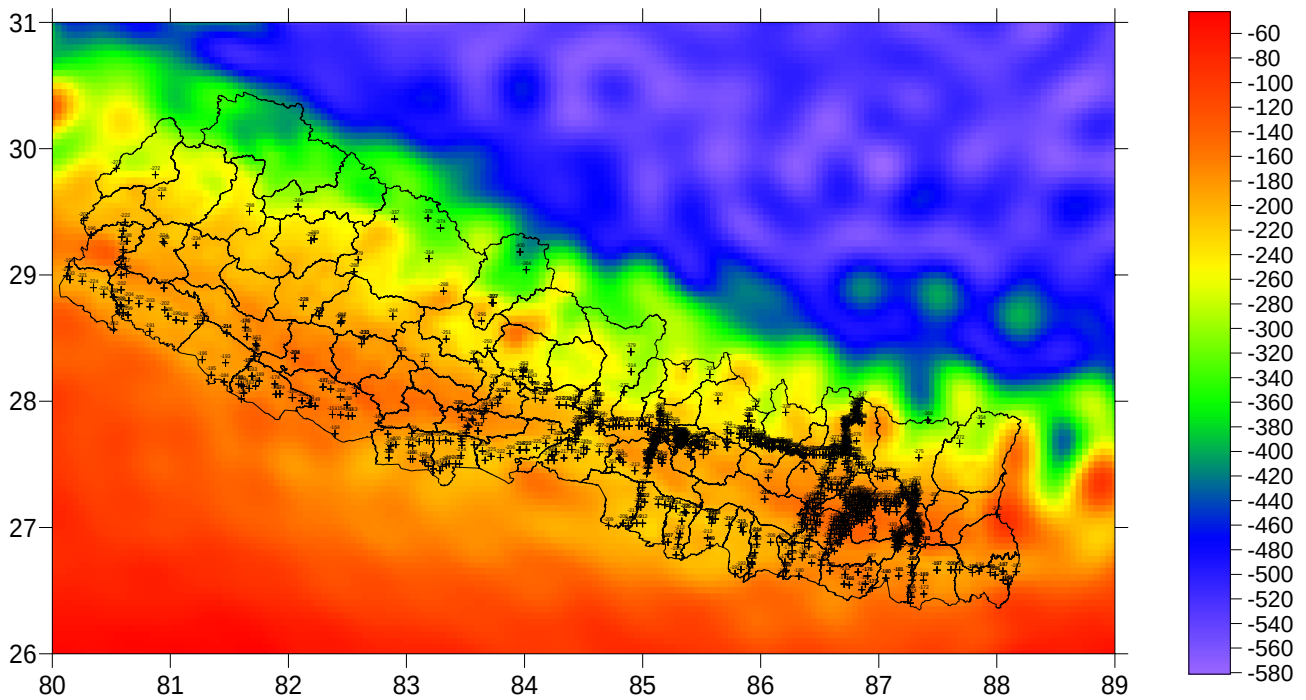


Fig. 5. Bouguer anomalies in Nepal from airborne and EGM-GOCE, with location of surface data. The process of error detection was aided by the zoom-function in “*surfer*” for the QC.

Terrain models and data reduction

The directory TC-RD are used for all terrain files, and the reduction of the airborne data for terrain and SRTM.

SRTM 15” data has been used for this geoid computation, averaged from the basic 3” data. Because the used 3” SRTM data had many voids in the high mountains, a revised file was made by interpolation of gaps from 30” data. *Basic file: srtm-15s-no9999.gri.*

Two support files are needed: 1.5’ x 1.5’ and 3’ x 3’ averages and a *mean height surface* for RTM. These files are *srtm30-05.gri* and *srtm-ref.gri*, made with *select* and *gfilt.job*.

All RTM terrain effects at altitude are computed by *TC*. Because airborne gravity data are filtered, the terrain effects must also be filtered correspondingly. The program *convolve* filters a data file, taking indirectly into account the varying speeds of the flight. The filter used is *filter.out*.

The job *rd_air* does the reduction of the airborne gravity data, namely

- 1) subtract EGM08 at altitude, sandwich grid interpolation (*geoip*)
- 2) compute terrain effects (*tc*)
- 3) filter terrain effects (*convolve*)
- 4) subtract terrain effects from EGM-reduced data (*geoip*)
- 5) final reduced file *airborne.rd* is obtained by thinning to every 3rd data point

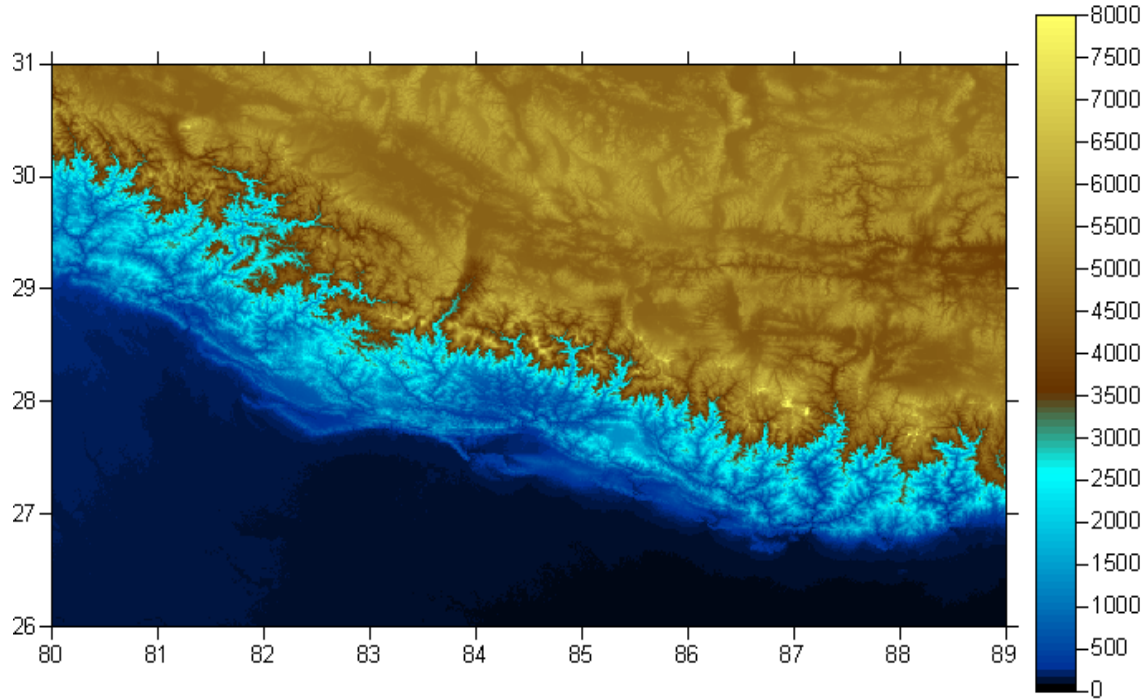


Fig. 6 SRTM-heights from 3" SRTM data with 30" fill-ins

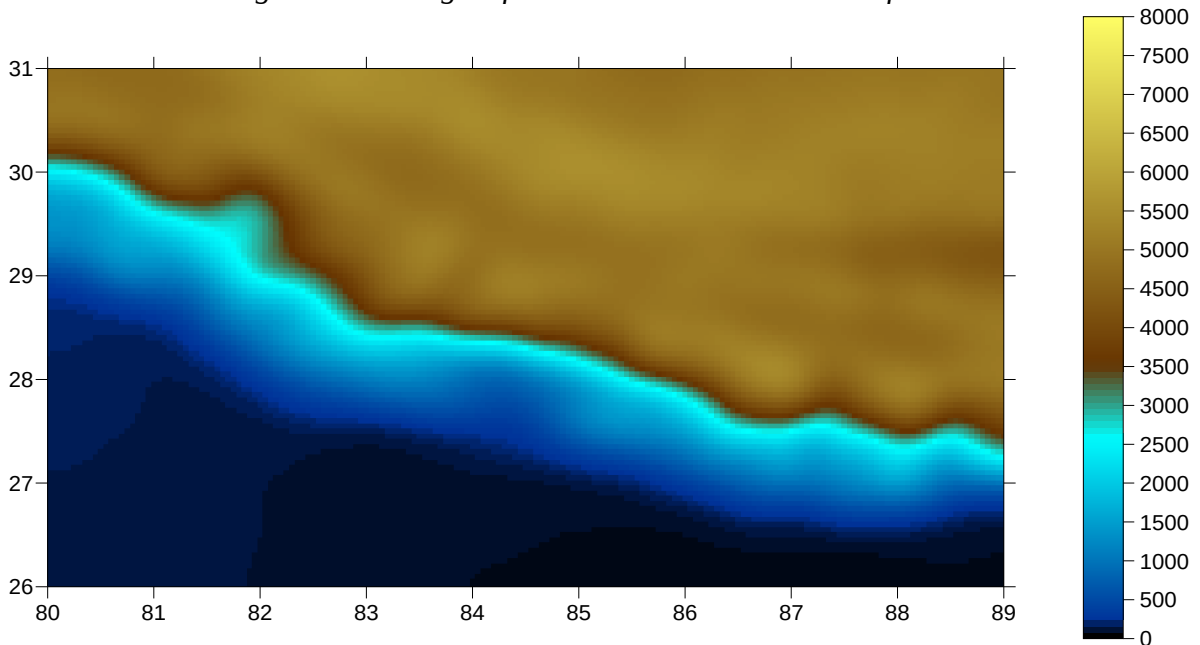


Fig. 7. Mean elevation surface used in RTM reduction (30' resolution filtered surface)

The overall smoothing of the EGM08-GOCE and RTM terrain reduction is shown below. It is seen that there is a rather large bias in the airborne data reduction. This is a serious problem for the geoid

determination, and for the final geoid (“*geoid3a*”) this bias has been removed. The source of this bias is currently being investigated, and might be related to the scale factor of S38.

Table 1. Statistics of data reductions.

Unit: mgal	Mean	Std.dev.
Airborne data, unreduced (4216 points @ 30 sec)	-13.8	119.8
Airborne minus EGM08/GOCE	2.6	31.6
Airborne – EGM08/GOCE – RTM	5.1	26.4
Surface free-air anomalies (549 points, @ 3’ res.)	-115.7	85.7
Surface free-air minus EM08/GOCE	-28.0	31.6
Surface – EGM08/GOCE – RTM	1.3	36.0

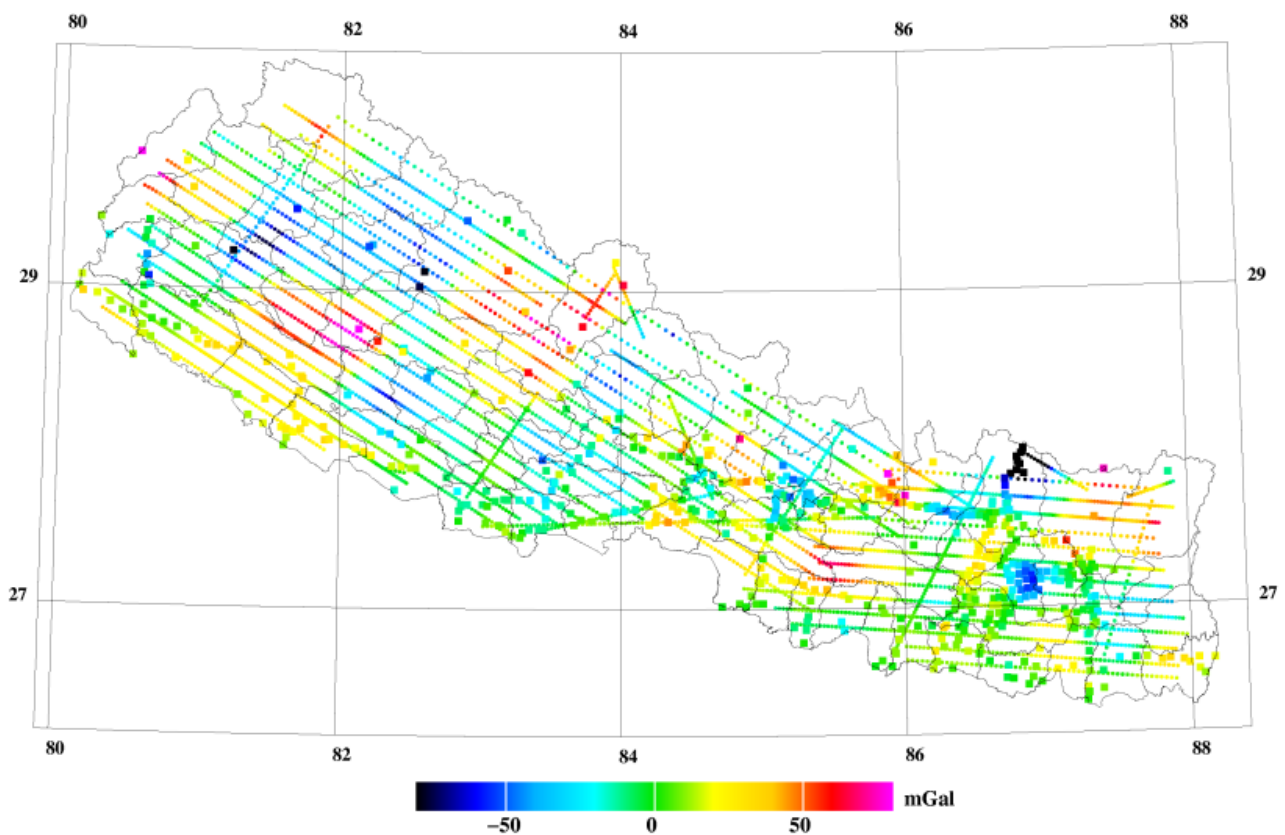


Fig. 7. Reduced airborne and surface data used for the geoid (“airborne.rd” and “surface.rd”)

The geoid computation

The geoid is obtained by several steps:

- 1) Downward continuation of the airborne data from height to the terrain level. This is done by least-squares collocation (*gpcol1a.job*). Note that this job should also include surface gravity, when available. To initiate collocation a covariance function must be fitted, see *gpfit.job*. The selection of the right covariance parameters is an art! Used values are $C_0 = 27$ mgal, $D = 5$ km, $T = 25$ km, based on joint fitting of surface and airborne data. The downward continuation results is in *downrd1.gri*.

- 2) The downward continued gravity on the geoid is converted to quasi-geoid heights at terrain-level (ζ) by spherical FFT, using a modified Stokes kernel keeping the GRACE information in EGM08 at degrees lower than 80, and a full downward/upward continuation as an intermediate step.
- 3) The terrain effects n_rtm are added (see “ $tc_n.job$ ” in TC-RD directory)
- 4) EGM08 effects $egm08n_720..gri$ is added. This yields the quasigeoid at sea-level ($quasigeoid.gri$)
- 5) The final gravimetric geoid is obtain by adding the correction $N-\zeta^*$ yielding the final “classical” geoid (“ $geoid3a.gri$ ”). This yields a geoid consistent with Helmert orthometric heights, cf. Heiskanen and Moritz, 1967, chap. 3.

Step 1) is $gpcol.job$ in directory *DOWNCONT* (NB: heavy computation, be patient). Steps 2-4) are in $geoid.job$, and step 5 in “ $n-zeta.job$ ”. The geoid can be interpolated to individual points by $geoip$ or $grid_int$ (nicer user interface).

Table 2. Statistics of geoid “restore” steps

Unit: meter	Mean	Std.dev.	Min	Max
Reduced geoid grid data (after FFT)	0.05	0.46	-2.73	2.36
Terrain geoid signal	0.04	0.48	-1.41	2.01
Conversion of quasigeoid to geoid	-1.18	1.15	-3.45	0.00
Final geoid	-42.76	16.39	-69.02	-20.59

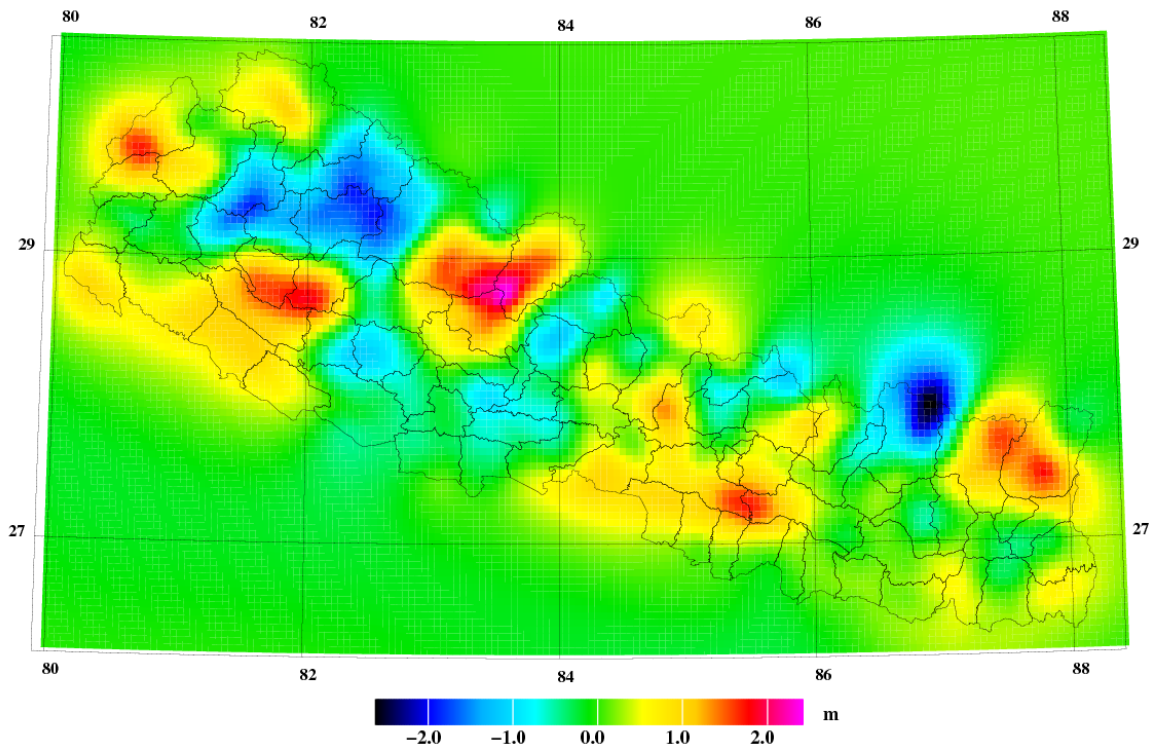


Fig. 8. Reduced geoid, before “restore” steps (“ $n_rd.gri$ ”)

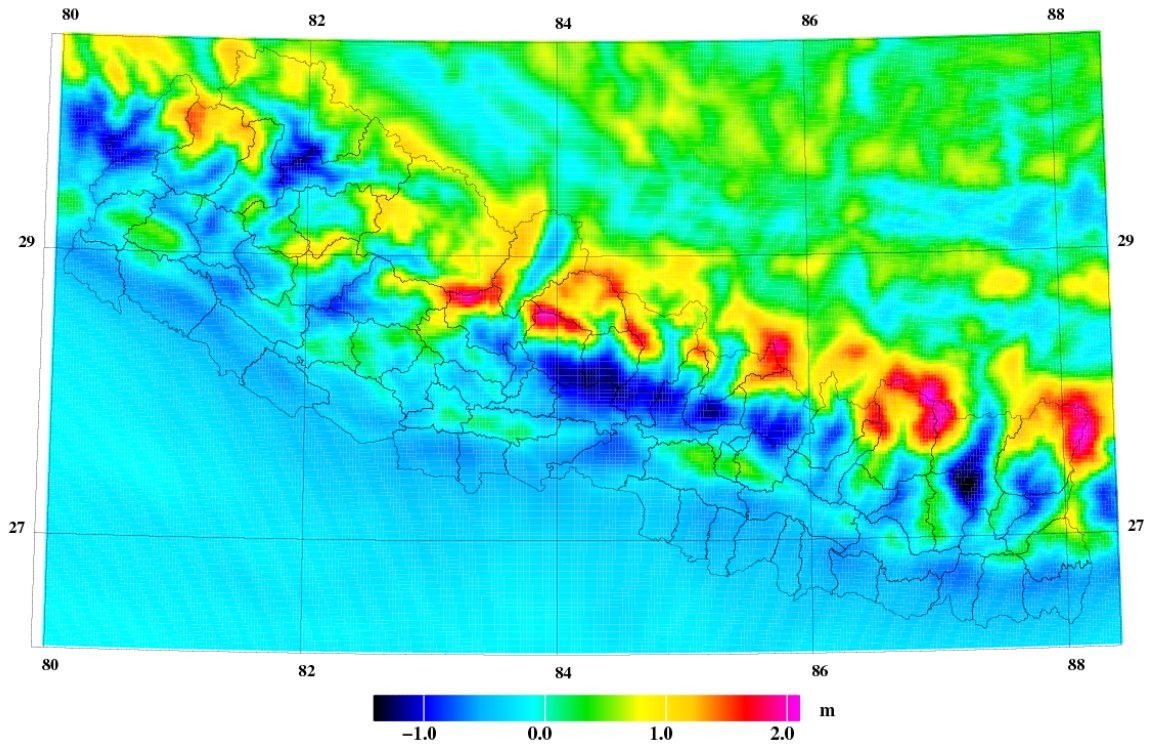


Fig. 9. Geoid terrain effects, computed by prism intergration (“n_rtm.gri”).

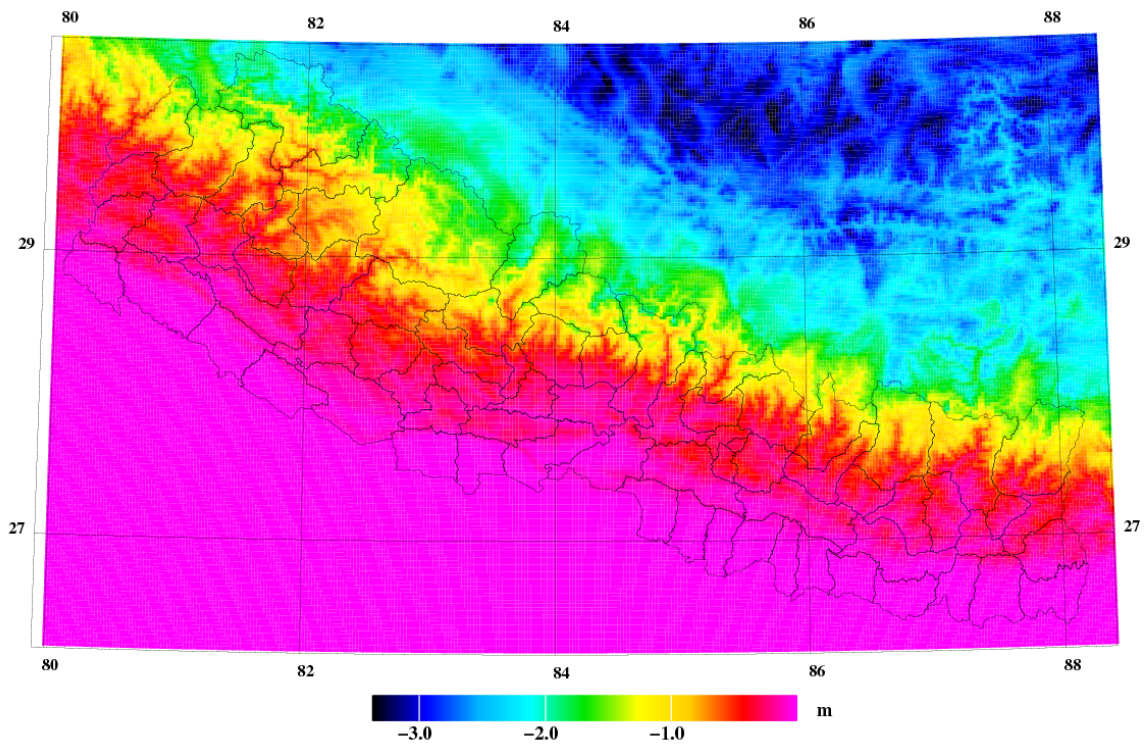


Fig. 10. Difference between geoid and quasigeoid ζ . The correction is largest at Mt. Everest

The final geoid is shown in Fig. 11-12. It should be noted that this geoid refers to a world height system, and for use GPS coordinates must refer to WGS84/ITRF.

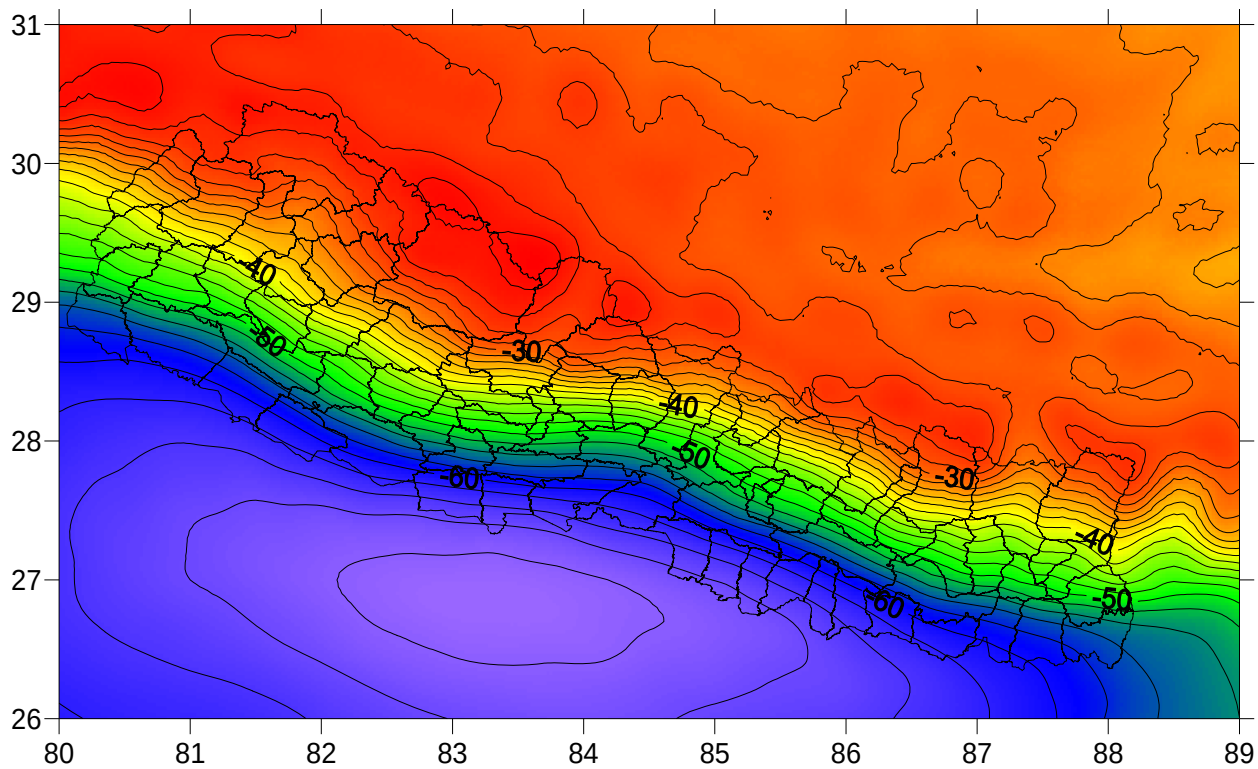


Fig. 11. Final geoid of Nepal (“geoid3a.gri” or “nepal_geoid_oct2011”). 2 m contours.

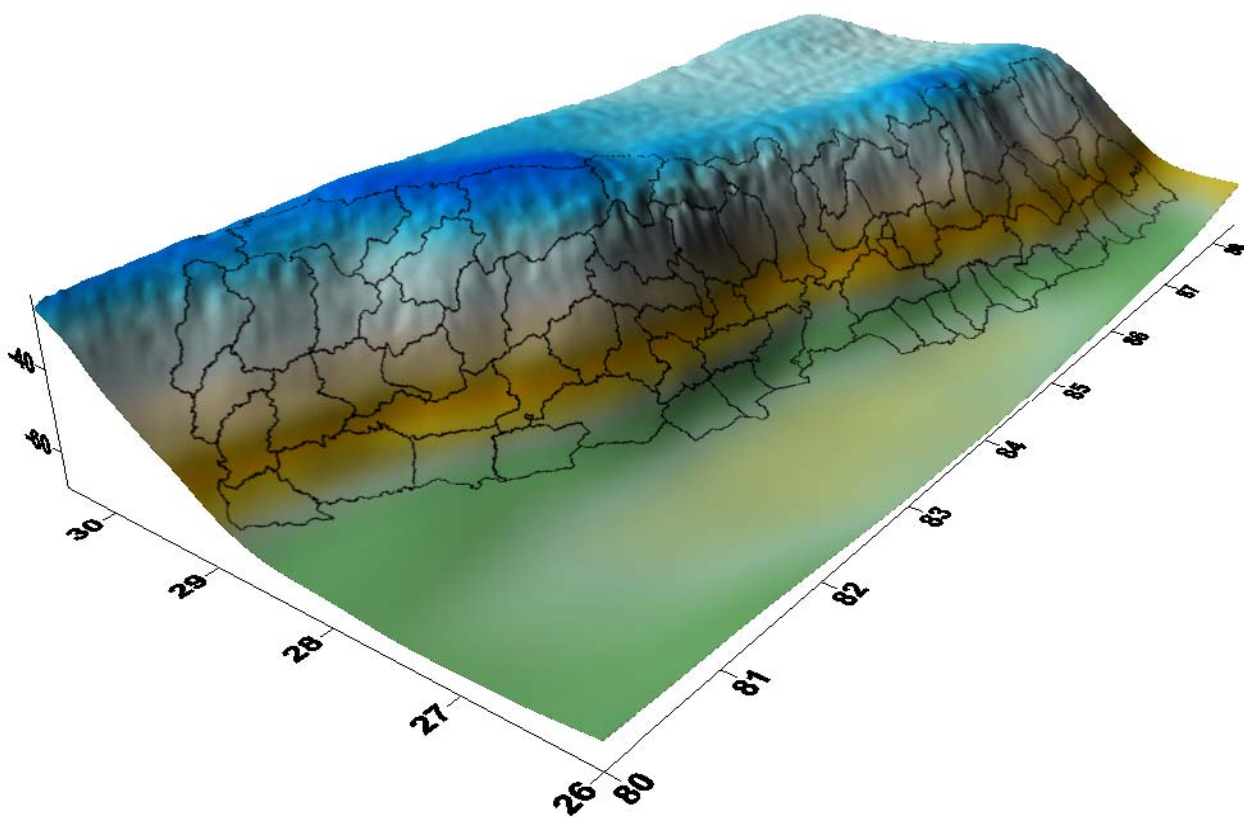


Fig. 12. Geoid of Nepal, 3D-view.

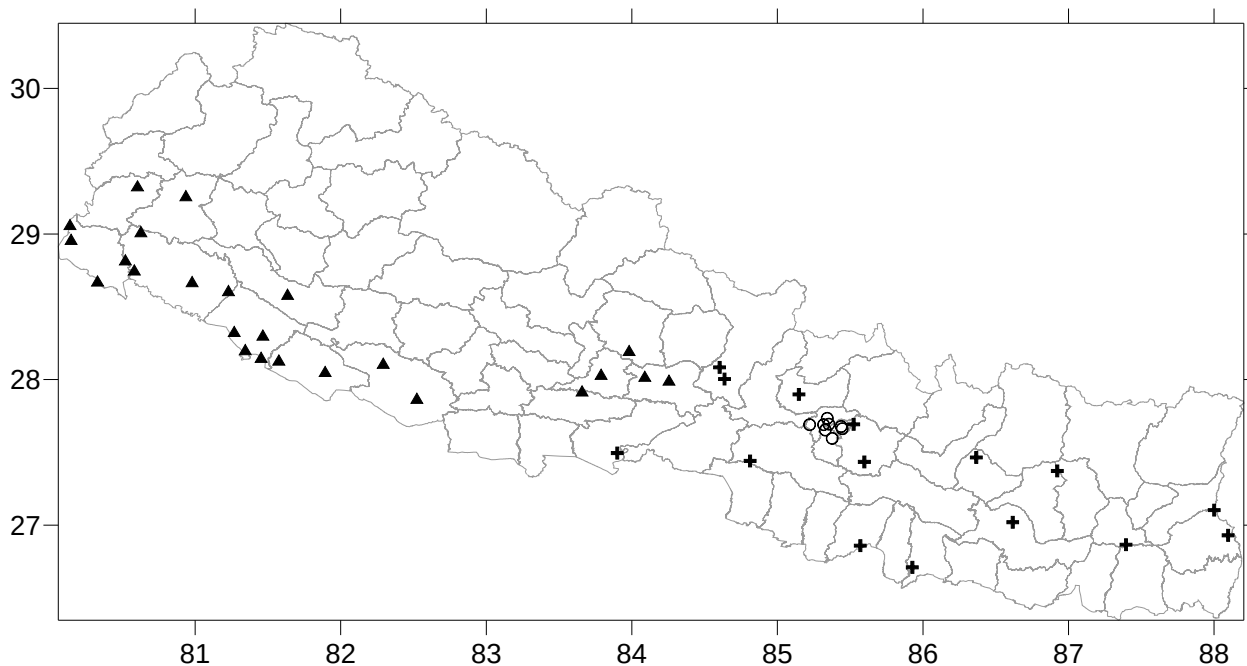


Fig. 13. Location of some GPS-levelling data in Nepal (triangle: “western”, cross: “eastern”, open circles “Kathmandu Valley”)

Comparison to 3 GPS-levelling campaigns outlined in Table 3. The GPS is not based on WGS84 (but local heights), and there are many outliers, especially in the eastern set. However, it clearly shows a geoid improvement over EGM08.

Table 3. Comparison to GPS-levelling (some obvious and very large outliers removed)

Unit: meter	Nepal geoid		EGM08 to n=2190	
	Mean	Std.dev.	Mean	Std.dev.
Kathmandu Valley (8 pts)	19.18	0.12	18.22	0.35
Eastern (15 pts, unedited)	48.78	4.50	48.53	5.18
Western (20 pts, large outliers removed)	19.73	0.36	19.25	0.52

To improve the GPS network of Nepal, the Nagarkot observatory datum should be transferred to all the Nepal networks. A test at DTU-Space of a few days of GPS Nagarkot data at AUSPOS processing site gave a new refined value of the fundamental GPS datum point in ITRF05. Table 4 below shows the fundamental data for Nagarkot GPS (27 41 33.8°N, 85 31 16.3°W).

Table 4. Nagarkot fundamental datum point height data

GPS ellipsoidal height, AUSPOS	2106.611 m
Antenna height correction	-1.142 m
Ellipsoid height of GPS ground point	2105.502 m
Earlier determined ellipsoidal height of Nagarkot (Univ. of Colorado)	2105.196 m
Levelled height, Nepal datum	2151.78 m
Geoid height from GPS and levelling (local datum)	-46.28 m
Geoid height from new model (global datum)	-47.72 m